around the hole, also begin parallel. But in this case, the warping drives them apart.

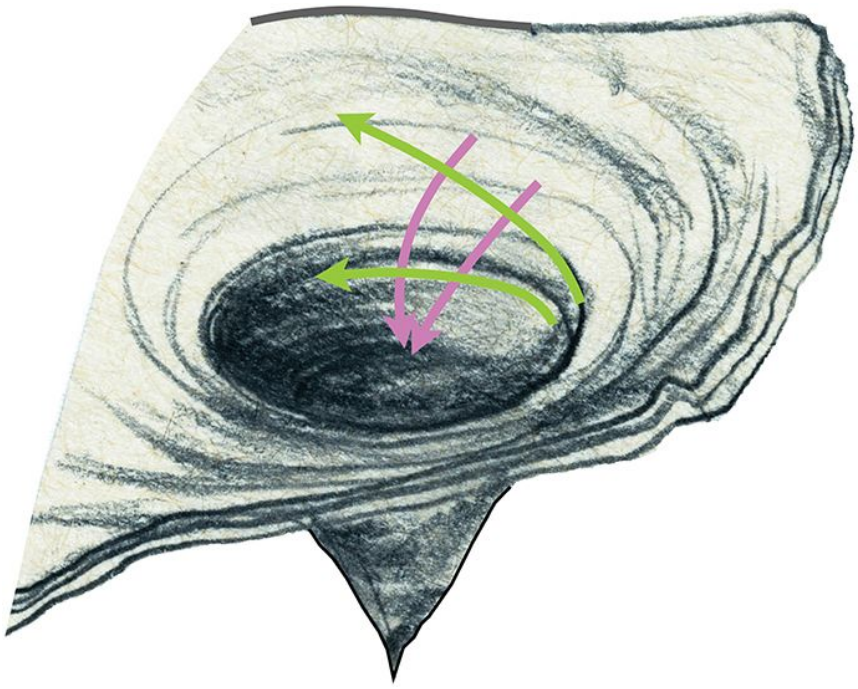


Fig. 4.6. Four paths for planetary motion in the vicinity of a black hole. The picture of the hole is extracted from Lia Halloran's drawing, Figure 4.5.

Several years ago, my students and I discovered a new point of view about these planetary paths. In Einstein's relativity theory there is a mathematical quantity called the Riemann tensor. It describes the details of the warping of space and time. We found, hidden in the mathematics of this Riemann tensor, lines of force that squeeze some planetary paths together and stretch others apart. "Tendex lines," my student David Nichols dubbed them, from the Latin word *tendere* meaning "to stretch."

Figure 4.7 shows several of these tendex lines around the black hole of Figure 4.6. The green paths begin, on their right ends, parallel to each other, and then the red tendex lines stretch them apart. I draw a woman